8 **(a)** A binary star consists of two stars that orbit about a fixed point C, as shown in Fig. 8.1.

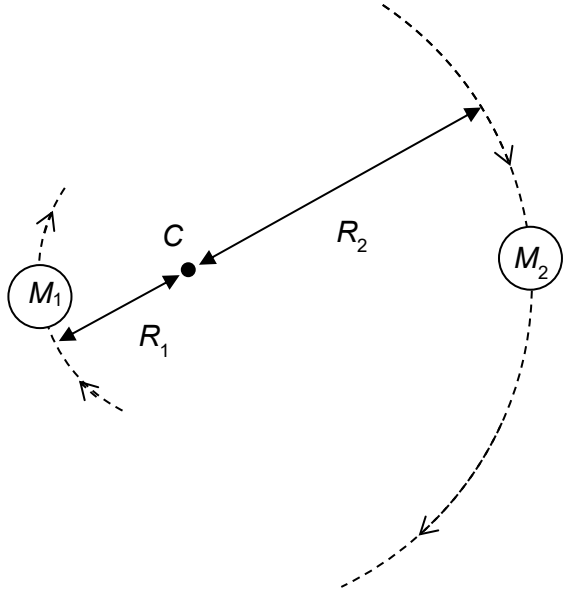

Fig. 8.1

The star of mass M_1 has a circular orbit of radius R_1 and the star of mass M_2 has a circular orbit of radius R_2. Both stars have the same angular speed ω about C.

(i) State the formula, in terms of G, M_1, M_2, R_1, R_2 and ω for

1. The gravitational force between the two stars

..... [1]

2. The centripetal force on the star of mass M_1.

..... [1]

(ii) The stars orbit each other in a time of 1.26×10^8 s. Calculate the angular speed ω for each star.

ω = ... rad s^{-1} [1]

(iii) Show that the ratio of the masses of the stars is given by the expression $\frac{M_1}{M_2} = \frac{R_2}{R_1}$.

[1]

(iv) The ratio $\frac{M_1}{M_2} = \frac{R_2}{R_1}$ is equal to 3.0 and the separation of the stars is 3.2×10^{11} m. Determine the radii R_1 and R_2.

R_1 = ... m

R_2 = ... m [1]

(v) By considering the expressions in **(i)** and using the data calculated in **(ii)** and **(iv)**, determine M_2.

M_2 = ...kg [3]

(b) Fig. 8.2 shows an electron entering a region between two oppositely-charged parallel metal plates. The plates have length 5.1 cm.

The electric field in the region between the plates is uniform and is zero outside this region.

The original direction of motion of the electron is normal to the electric field.

The original speed of the electron is $v = 1.7 \times 10^7$ m s^{-1}.

The electric field strength between the plates E is 4000 V m^{-1}.

The electron exits the plates at an angle θ to the horizontal.

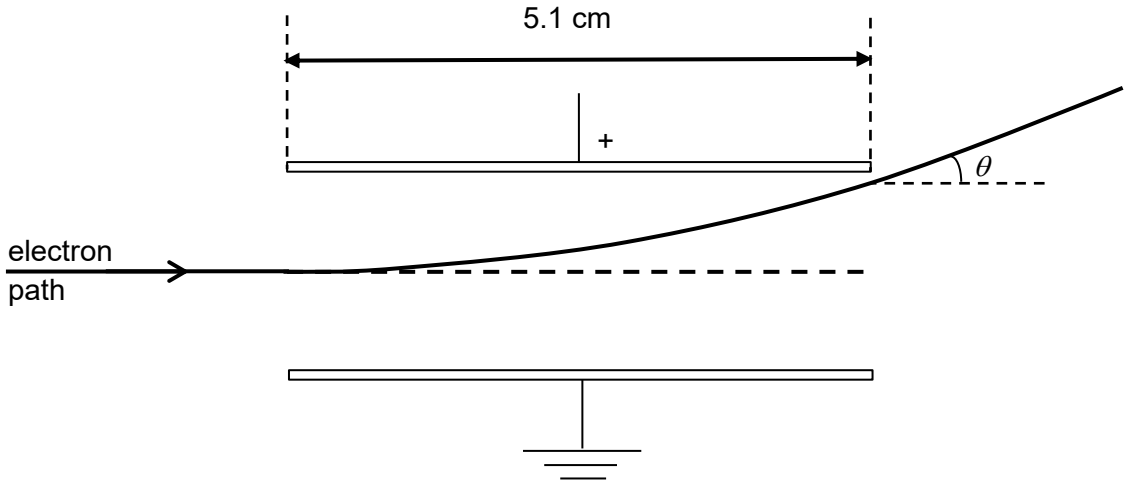

Fig. 8.2

(i) Show that the acceleration of the electron inside the electric field is 7.0×10^{14} m s^{-2}.

[1]

(ii) Calculate the magnitude of the final velocity of the electron, and the angle θ.

final velocity = ... m s^{-1}

θ = ... ° [4]

(iii) A proton is projected with the same initial velocity along the same line. Without detailed calculation, draw the path that the proton takes on Fig. 8.2. Explain your answer.

.....

.....

..... [2]

(c) Fig. 8.3 shows a uniform magnetic field *B* denoted by the shaded area. An electron moves into the field at the same speed *v* as in **(b)**, and is also deflected from its original path. The original direction of motion of the electron is normal to the magnetic field.

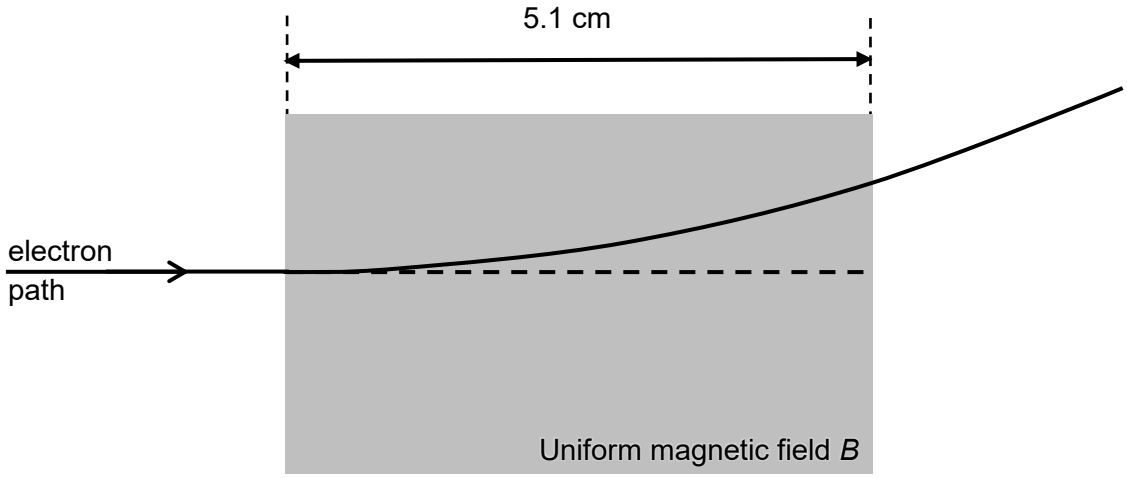

Fig. 8.3

(i) State the difference between the shape of the path taken by the electron in the magnetic field, and the shape of the path taken by the electron in the electric field described in **(b)**. Explain this difference.

.....

.....

.....

.....

.....

.....

..... [3]

(ii) State and explain how the final speed of the electron after passing through the magnetic field compares with the final speed of the electron after passing through the electric field in **(b)**.

.....

.....

..... [2]